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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-1

Introduction to the EV Charging System

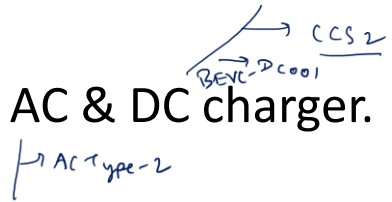
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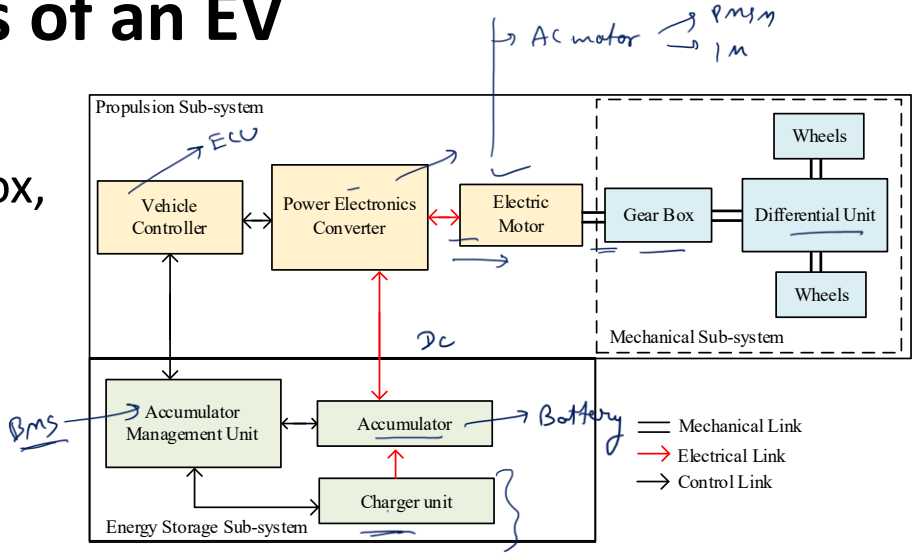
Course Introduction

- Covers topic like
  - Introduction to EV charging system, building blocks of an AC and DC charger.
  - Types, Operating principles, design, and control of power conversion unit (**AC-DC** and **isolated DC-DC** power converter).
  - Communication aspects.
  - Charging procedures and protocols: AC & DC charger.



Subsystems of an EV

- EV consist of following sub-systems
  - Mechanical sub-system: gearbox, differential, mechanical linkages
  - Electric motor
  - Power electronics converter
  - Vehicle Controller
  - Energy Storage sub-system (Source)
  - Vehicle Communication



Key EV Battery Information (INDIA)

Vehicle Segment	Battery Capacity	Battery Voltage
E-2W	1.2-8 kWh	48-72V
E-3W Passenger/goods	3.6-8 kWh	48-72V
E-car with LV powertrain	21 kWh	72V
E-car with HV powertrain	30-80 kWh	350-500V

Low voltage

40 - 120V

75% - 115%

HV (350 - 500V)

Data: https://www.niti.gov.in/sites/default/files/2021-08/HandbookforEVChargingInfrastructureImplementation081221.pdf

EV Battery Charging

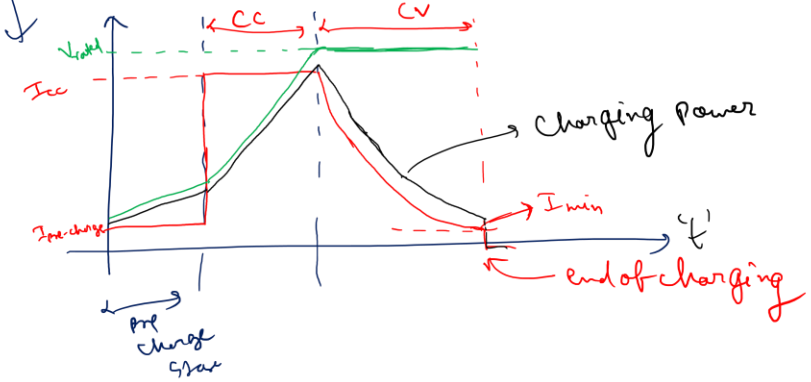
- EV Battery charging can be done in the following ways:
  - Conductive charging: AC or DC chargers
  - Contactless charging: inductive, capacitive → Static
  - Dynamic contactless (on-road vehicle charging) charging → increased range
  - Battery swapping → range is increased to ∞ (if the battery swapping stations are placed in near by distance)
    → stoppage time is minimal as compared to Conductive Charging

} range anxiety

Battery Charging Modes

2.7V | 3.6V | 4.2V → 115%
75% 35V | 48V | 55V

- Key battery charging modes are:
  - CC (constant current) → Charging is done with constant current → overheating; → monitoring of v/g of battery is required
  - CV (constant voltage) → Charging is done with constant voltage → overheating; → over current
  - CCCV (constant current followed by constant voltage) → CC → fast charging, CV → time taking Part of charging
  - CPCV (constant power followed by constant voltage) → Charging the battery with high currents at low SOC's and the current goes down as charging progress



- Multistep CC-CV
- Pulse charging
- Variable current charging

CC-CV

Need of an EV charger

- The requirement of charging profile is determined by the battery management system (BMS).

- EV charger needs to support specific charging profile with required protection, safety, and communication (for monitoring and metering).

  - Charging is done from AC grid or uncontrolled DC source (from renewables)
  - Battery which is the DC source

Building Blocks of an EV Charger

- Basic building blocks of an EV charger
  - Power Conversion unit
    - AC-DC converter, isolated DC-DC converter, Filters
  - Communication Ports
    - Proximity pilot, control pilot, CAN H, CAN L, PLC modem
  - EVSE (electric vehicle supply equipment): AC, DC EVSE
  - Connectors and cables:
    - Charging plug from EVSE, Vehicle inlet connector
    - Cables → 5m

Thank You